Reasoning Over Building System Graphs

I. Use Case Description	
Use Case Name	*CareAir: Explainable Reasoning for Healthcare Ventilation Compliance*
Use Case Identifier	*HVCC-10*
Source	*Center for Architecture Science and Ecology, Rensselaer Polytechnic Institute*
Point of Contact	*Ahosan Habib, habiba5@rpi.edu* *Nipun Deelaka, pathin@rpi.edu* *Andy Cheng, chengy11@rpi.edu* *Carina Liu, liuc17@rpi.edu*
Creation / Revision Date	*11/06/2026 (Version 10)*
Associated Documents	- *ANSI/ASHRAE/ASHE Standard 170-2025, Ventilation of Health Care Facilities* - *CDC, Guidelines for Environmental Infection Control in Health-Care Facilities -the open source of the numeric thresholds actually encoded.* - *ASHRAE Standard 223P, Semantic Data Model for Analytics and Automation Applications in Buildings* - *DOE / PNNL Prototype Building Models - Hospital, New York variant (EnergyPlus).* - *Prior OE course project: indoor-environment-manager (OE 2022), Chung, Rook and Jacoby-Cooper - reviewed for reuse (Section X)*

II. Use Case Summary	
Goal	*Enable a hospital infection preventionist or facilities engineer to determine whether critical care spaces currently satisfy the ventilation requirements that their clinical designation imposes; to identify which air-handling equipment is responsible when a space falls out of compliance or would fall out of compliance during a planned outage; and to keep both the determination and its history correct when a space is re-designated to a different clinical use.*

Requirements	*R1. Represent the clinical function of a space as a ROLE the space plays over a period, not as a permanent subclass of the space. Ventilation requirements follow clinical function and hospital spaces are re-designated in service.* *R2. Represent ventilation requirements declaratively - pressure relationship and minimum air change rate - as data associated with a clinical designation, carrying units, a staleness window and a citation to the published source of the threshold, rather than hard-coded in application logic.* *R3. Represent the air distribution topology - which equipment connects to what, through which connections, carrying which medium - so that the relation between an air handling unit and the spaces it serves is derived from the components each system contains and the rooms they connect to, rather than asserted.* *R4. Associate sensor measurements with the sensor that produced them and, through it, the room they were recorded in, carrying a value, a unit and a time.* *R5. Produce compliance determinations by query and inference over the model, with no space-specific procedural code, so that applying a framework to a new room, or adding a component or sensor, changes what is evaluated with no change to code or queries.* *R6. Retain provenance for every determination: which observation, at what time, compared against which requirement, and the published source from which the threshold was taken.* *R7. Identify the building service systems and components that may have contributed to a failed assessment, as candidates linked to the failing measurement, without asserting a confirmed cause.* *R8. Operate read-only. The system reports and explains; it never actuates equipment and must never become a dependency of a life-safety control loop.* *R9. Log sensor measurements at request. A measurement is taken and recorded when a request asks for one; the system does not poll sensors by itself, and every answer rests only on the measurements logged.*
Scope	*In scope for this assignment:* *• Two clinical designations: airborne infection isolation and operating room.* *• The CDC compliance framework and its policies, which are minimum-bounded, maximum-bounded or range-bounded over a measured value.* *• Designation as a time-bounded role, and the re-designation of a space from one clinical use to another.* *• Building service systems (HVAC in particular), their components and sub-components, the sensors among them, and the rooms they serve.* *Out of scope, deferred as term extensions:* *• Quantification of energy savings and the penalty of over-ventilation excluded because it requires a thermal-energy model and an energy price signal, and the compliance question does not depend on either: a space either meets its pressure and air-change requirement or it does not, whatever that costs to achieve.* *• Fault detection and diagnosis, and control actions excluded by requirement R8, which keeps the system read-only. Diagnosis asserts a cause; this project asserts only whether a requirement is met, which evidence supports that, and which components are candidates linked to that evidence; it never names a confirmed cause. Crossing into actuation would also make the system a dependency of a life-safety control loop,*

	which R8 forbids. *• Water, medical gas and electrical systems are excluded because each carries its own requirement vocabulary and none bears on the ventilation measurements in scope, such as air change rate and pressure. Including them would enlarge the term list without exercising any new reasoning.* *• Patient data of any kind. Excluded on both ethical and modelling grounds: no competency question requires it, and admitting protected health information would place the model under HIPAA for no analytical gain. The model describes spaces and equipment only.* *• Selecting among jurisdictional variants excluded because choosing between variants requires a model of legal authority - which code is adopted in which jurisdiction, as of when - which is a separate problem. Pre-condition P6 instead fixes a single requirement set as the evaluation context, so the reasoning can be demonstrated without that machinery.* *• Every trigger except T1. Excluded because T2 to T4 each require running infrastructure - a scheduler, a live observation stream, a change-detection service - that this project is not building. Each is documented in the Triggers row because it shapes the design, but only T1, a user-initiated request, is implemented.*
Priority	*High within the project.*
Stakeholders	***Primary**: healthcare administrators and nurses, who are accountable for the ventilation compliance of critical care rooms; system maintenance technicians, who service the systems and sensors that serve each room; and system administrators and facilities engineers, who own the HVAC systems.* ***Secondary**: surgical and nursing staff occupying the affected spaces; the hospital compliance officer preparing for Joint Commission and CMS survey; patients, particularly immunocompromised and airborne-isolation patients, who bear the clinical consequence of a containment failure; hospital administration, carrying regulatory and financial risk.*
Description	*Building codes are written for people to read. There is no machine-checkable form of them, so compliance is established by a person walking the building with a clipboard and the reasoning behind a determination is not recorded anywhere. This use case takes the sharpest instance of it - hospital ventilation, where the consequence of being wrong is clinical rather than financial - and builds the smallest version of it that can be completed.* *Hospitals are the one building type in which ventilation performance is a clinical matter rather than only a comfort or energy matter. ASHRAE Standard 170 assigns each space a required pressure relationship and a minimum air change rate according to what clinically happens in it: an airborne infection isolation room must be held negative with respect to the corridor so that contaminated air cannot escape, while an operating room must be held positive so that unfiltered air cannot enter.* *The requirement follows the clinical function of the space. The ability to satisfy it follows the physical air distribution topology. The evidence that it is satisfied comes from sensor measurements logged at request. These three bodies of knowledge live in three unconnected places: the clinical space program, the mechanical model and the building management system. A compliance question cannot be answered without joining all three.* *This use case proposes an ontology that joins them, so that the question is asked once and answered by inference. The principal actors are the system maintenance technician, the healthcare administrator or nurse, and the*

	system administrator. The system reports which systems and sensor readings relate to a room, reports compliance state, and explains a failure by naming the policy that was not satisfied, the measurement compared with it, and the HVAC systems and components that may be involved. *One claim underlies the whole use case. That a particular air handling unit serves a particular room is not declared in any system of record - it is a consequence of how the ductwork was built. It must be derived from the components each system contains and the rooms those components connect to, and the determination that follows must carry its own justification: the observation it rests on, the threshold it was compared against and the policy that sets that threshold. Deriving the relation and justifying the answer are what separate this from a database query and they are why the problem needs an ontology rather than a schema.*
Actors / Interfaces	*PRIMARY (invoke the use case and benefit from the result):* *• System maintenance technician - asks which systems serve a room and what its sensors have recorded.* *• Healthcare administrator or nurse - asks which rooms met the CDC ventilation standard and why a room did not.* *• System administrator - asks which HVAC systems and components relate to rooms that failed.* *SECONDARY (acted upon by the system):* *• 223P knowledge graph - spaces, equipment, connections, media, observable properties.* *• Compliance framework - the CDC framework, its policies and the rooms it applies to.* *• Compliance policies - lower and upper bounds with units.* *• Building management system / BACnet and IoT sensor API - differential pressure and airflow observations.* *• IFC-LD building model - source of spaces, storeys and containment relationships.* *• SPARQL endpoint and SHACL validation engine - evaluate constraints and return results.*
Pre-conditions	*P1. A 223P model of the nursing unit exists and has passed SHACL validation.* *P2. Every space within scope carries a clinical space type classification. A space with no clinical type cannot be evaluated and must be reported as such rather than skipped.* *P3. Each building service system is linked to its components, including nested sub-components, and each component to the rooms it connects to, so that the path from the air handling unit to the rooms served can be followed.* *P4. Sensors are connected to the rooms they monitor. Each sensor measurement carries a value, a unit and a time, and is logged at request; the system does not poll sensors.* *P5. Compliance policies of the in-scope framework are encoded as data with a lower and/or upper bound and a unit, and each compliance assessment is linked to the measurements it used.* *P6. A single requirement set is fixed as the evaluation context.* *P7. The requesting user is authenticated and authorised to view facility data.* *P8. Each component is linked to the measurements it acts on, and each derived measurement is linked to the measurements it was derived from.*
Post-conditions	*Q1. Every in-scope space carries a compliance determination of compliant, non-compliant or indeterminate.*

	Q2. Each determination carries the observation and timestamp that produced it, the threshold compared against, and the policy and source from which that threshold derives, so the result can be reproduced and audited. *Q3. Indeterminate results - arising from a missing sensor, a stale observation, or an unclassified space - are reported explicitly and are never silently treated as compliant. The system fails visibly rather than optimistically.* *Q4. For a failed room, the HVAC systems serving it and the candidate components are returned together with the policy and the measurement that point to each.* *Q5. For a room query, the systems serving the room and the sensor measurements recorded in it are returned, each measurement with its value, unit, time and sensor.*
Triggers	*T1 is the only trigger in scope for this project, on review feedback. T2 through T4 are documented because each shapes the design, but each also requires running infrastructure - a scheduler, a live stream - that this project is not building.* *T1. IN SCOPE. User-initiated (external). A preventionist requests current compliance state, typically before a patient placement decision; an engineer submits a planned-outage query.* *T2. DEFERRED. Scheduled (temporal). Periodic re-evaluation maintains a standing compliance state.* *T3. DEFERRED. Measurement-driven (internal). A logged measurement crossing the bound of a policy changes the compliance state of a room.* *T4. DEFERRED. Model-change (internal). Reclassification of a space, modification of the topology or adoption of a new edition of the standard invalidates existing determinations and triggers re-evaluation. Because requirements are inherited from space type rather than attached per room, a reclassification alone is sufficient to change what is checked.*
Performance Requirements	*Standing compliance state returned interactively, target under five seconds for a nursing unit; tracing failed rooms to HVAC systems and components is a diagnostic query, target under ten seconds.* *Measurement time is set at request, under T1. Sensor measurements are logged when a request asks for one, so the system holds no standing compliance state between requests, and this revision specifies no re-evaluation interval. Each assessment is linked to the measurements it used, so a determination always refers to the time those measurements were taken.* *Sizing: one nursing unit is on the order of thirty spaces, one AHU, forty terminal units, one hundred points; the representation must extend to a full campus (~10,000 spaces) without redesign.* *Maintainability: thresholds, units, and citations are data in the graph, so a change of adopted edition is a data update rather than a code change. The system is advisory and read-only and must never become a dependency of a life-safety control loop.*
Assumptions	*Sensor readings are trusted as reported; sensor fault detection is out of scope. Sensor measurements are logged at request, not collected by self-polling, so a measurement exists only for a time at which one was requested.A single requirement set is in force for the facility for the whole period modelled.* *For initial development the DOE Prototype Hospital supplies topology and zone volumes, with measurements derived from an EnergyPlus run and*

	logged at request. *DECISION (measurement logging), taken on review feedback. Sensor measurements are logged at request and not by self-polling. A measurement carries the time at which it was taken, and each compliance assessment is linked to the measurements it used, so a determination always refers to when its evidence was taken. A staleness window on requirements and an expected reporting frequency on sensors are not modelled in this revision, because the system does not poll; they are deferred.* *DECISION (extensibility), taken on review feedback. Where a modelling choice is open, this project resolves it in favour of later extension rather than present simplicity, because the team intends to carry the work past the end of the semester. Accordingly compliance policies are kept separate from the assessments that apply them, and measurements separate from the policies that evaluate them, so that a new policy, framework or sensor type can be added without changing existing assessments.*
Open Issues	*Terminology harmonisation across the team. The space-type labels in the extracted building data and the space types named in the requirement encoding do not currently match, and five people are about to write terms for five modules. This is the subject of Assignment 4 and is recorded here because it blocks module composition, not because it is unsolved in principle.*

III. Usage Scenarios

Usage scenario 1 - Locating the systems and sensor readings that relate to a room

Actor: *System maintenance technician (primary). The building knowledge graph of service systems, components and sensors (secondary).*
Goal: *Know which building service system serves a given room, and which sensor readings have been recorded in that room, so that a fault can be narrowed down before anyone opens the plant or the ceiling.*
Scenario: *A work order arrives for room 4E-12, reported as stuffy. Before inspecting anything, the technician asks which building service systems serve the room and what its sensors have recorded. Nothing in the data says which system serves 4E-12, so the system infers it. For each building service system it looks at the components the system contains and at the rooms those components connect to. HVAC system HVAC-3 contains the air handling unit AHU-3, the supply duct SD-3, the variable air volume box VAV-412 and the diffuser DF-412, and VAV-412 and DF-412 connect to room 4E-12, so the system concludes that HVAC-3 serves 4E-12. The same inference works for any kind of building service system and stays correct when a component is added or re-connected. The technician then asks for the readings recorded in the room. The sensors are not polled on a schedule; the system logs a measurement when a request asks for one, so the answer lists the readings logged so far, each with its value, unit, time and the sensor that produced it. The technician goes to the named system and sensor instead of searching the building. This use case starts from a room and a work order and involves no compliance result.*
Pre-conditions: *Rooms, building service systems and their components (sensors included) are described in the building model, and each component is connected to the rooms it relates to. Sensor measurements are logged at request, not collected by self-polling, so only logged measurements can be returned.*

Usage scenario 2 - Verifying the ventilation compliance of critical care rooms

Actor: *Healthcare administrator or nurse (primary). The CDC compliance framework and the symbolic compliance reasoner (secondary).*
Goal: *Know which rooms of a given type, for example ICU rooms, met the CDC ventilation standard at the*

time they were tested, and for any room that did not, which compliance policies were not satisfied and which measurements show it.
Scenario: *Ahead of an infection-control review, an administrator asks which ICU rooms were within the CDC ventilation standard at the time they were tested. A test is a request: the system logs the needed sensor measurements at that time, the reasoner compares them with the policies of the CDC framework, and the result is recorded as an assessment of the room. The system lists the rooms that satisfied every policy and those that did not. For a room that did not, the administrator asks why. The system names each policy that was not satisfied, for example the minimum outdoor air changes per hour, and the measurement that was compared with it, with its value and time. If a measurement needed by a policy could not be logged, that policy is reported as unknown, not as satisfied. The administrator can then decide whether the room stays out of use and hand facilities the evidence.*
Pre-conditions: *The CDC compliance framework and its policies, each with a lower and/or upper bound and a unit, are in the model and apply to the rooms in scope, ICU rooms among them. Sensor measurements for a test are logged at request. The symbolic reasoner evaluates each policy against its measurement and the result is recorded as a policy assessment of the room's compliance assessment.*

Usage scenario 3 - Tracing compliance failures to HVAC systems and components

Actor: *System administrator responsible for the building systems (primary). The building knowledge graph and the assessments from Use Case 2 (secondary).*
Goal: *Know which HVAC systems are related to the rooms that failed compliance, and which components of those systems may be causing the failure.*
Scenario: *After the failing ICU rooms from Use Case 2 are reported, the administrator asks which HVAC systems are related to them, and the system returns the systems that serve those rooms. The administrator then asks which components of those systems could be responsible. The system returns the components, including parts nested inside other parts, that act on the measurement that was not satisfied or on a measurement it was calculated from, together with the policy each is linked to. The administrator inspects these candidates first instead of the whole system. The system reports candidates and their evidence; a failed room is not proof of an HVAC fault, and the system does not diagnose or act on equipment.*
Pre-conditions: *Use Case 2 assessments exist. HVAC systems, their components and sub-components are described, each component is linked to the measurements it acts on, and each derived measurement is linked to the measurements it was derived from.*

How the use cases relate: *Use Cases 1 and 3 both use the relation between a room and the HVAC system that serves it, on purpose, but they are not the same use case. In Use Case 1 that relation is the starting question: it is asked for any room and any building service system, with no compliance involved, and it is joined with the room's sensor readings. In Use Case 3 it is only the first step, applied to the rooms that failed in Use Case 2; the use case then adds what Use Case 1 never does, a trace from the failing measurement back to candidate components. Use Case 3 therefore does not replace Use Case 1, because it returns neither the other service systems nor the room's readings. Use Case 2 supplies the failed rooms that Use Case 3 starts from.*

IV. Basic Flow of Events

The normal courses below are written one per use case. All three share the same shape - receive a request about a room or a set of rooms, work out what is not stored, gather the evidence, and return the answer with the evidence behind it - but they diverge in the middle, and each therefore has its own step table. In every case the sensor measurements are logged at request; none is collected by self-polling.
Use Case 1 needs no compliance policy: the system works out the serving systems from the building topology and the room's measurements from the sensors connected to it. Use Case 2 logs the measurements for the test, records whether each policy was satisfied by comparing the measurement with the bounds of the policy, and classes the room as satisfied, partially satisfied, unsatisfied or unknown. Use Case 3 starts from the assessments of Use Case 2 that were not satisfied and traces them back to the HVAC systems and components involved.

Use Case 3 therefore depends on Use Case 2 having produced assessments, and it reports candidate components and their evidence, not a confirmed cause.
In all three the final step returns the answer together with the evidence behind it, and in none of them does the system write to the building. The alternate flow in Section V applies to Use Case 2, and through it Use Case 3, because either can reach the point where a measurement is required and find it missing.

Basic / Normal Flow of Events: Use Case 1 - Locating the systems and sensor readings of a room			
Step	***Actor (Person)***	***Actor (System)***	***Description***
1	*System maintenance technician*	—	*Requests the building service systems and the sensor measurements for a named room.*
2	—	*Reasoner*	*Works out which systems serve the room, from the components each system contains and the rooms those components connect to.*
3	—	*Reasoner*	*Works out which measurements were recorded in the room, from the room each one was measured in and the sensor that produced it.*
4	—	*Observation store*	*Returns the measurements logged so far, each with value, unit, time and sensor,*
5	—	*User interface*	*Presents the serving systems, including HVAC, and the measurements.*
6	*System maintenance technician*	—	*Goes to the named system and sensor to inspect the fault.*

Basic / Normal Flow of Events: Use Case 2 - Verifying the ventilation compliance of critical care rooms			
Step	***Actor (Person)***	***Actor (System)***	***Description***
1	*Healthcare administrator or nurse*	—	*Asks which ICU rooms met the CDC ventilation standard at the time they were tested.*
2	—	*Compliance service*	*Retrieves the ICU rooms, which are the rooms the CDC framework applies to, and the policies of the framework.*
3	—	*Observation store*	*Logs, at this request, the sensor measurements that the policies need for each room, each with value, unit and time.*
4	—	*Reasoner*	*Compares each measurement with the lower and/or upper bound of its policy, in the same unit, and records each policy assessment as satisfied, unsatisfied or unknown.*
5	—	*Compliance service*	*Records the compliance assessment of each room, with its context and measurements, and classes it as satisfied, partially satisfied, unsatisfied or unknown from its policy assessments.*
6	—	*User interface*	*Lists the rooms that met the standard and, for any other room, each policy that was not satisfied with its bound and the measurement compared with it.*
7	*Healthcare administrator or nurse*	—	*Decides whether the room stays out of use and passes the evidence to facilities.*

Basic / Normal Flow of Events: Use Case 3 - Tracing compliance failures to HVAC systems and components			
Step	***Actor (Person)***	***Actor (System)***	***Description***
1	*System administrator*	—	*Asks which HVAC systems relate to the rooms that failed compliance.*

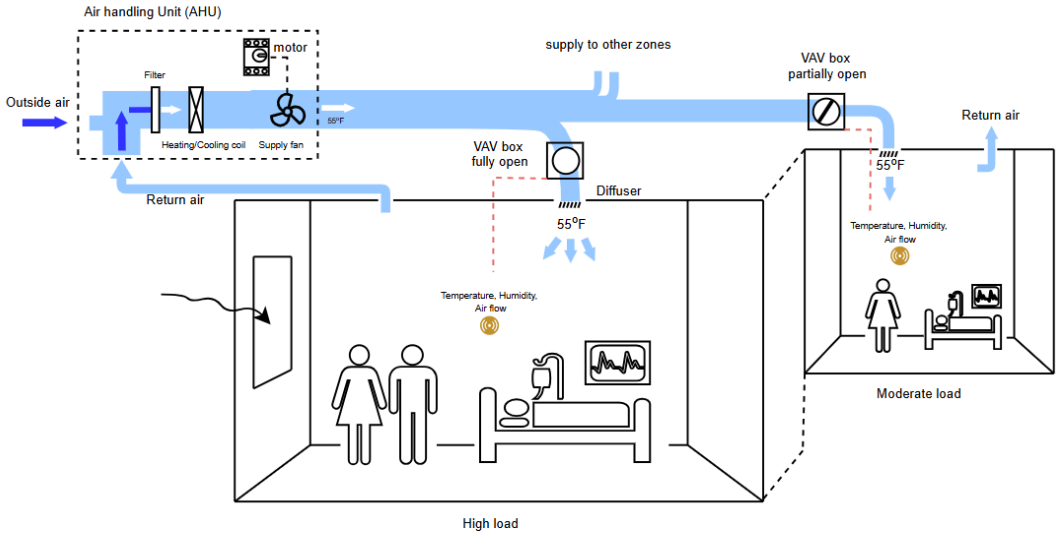
2	*—*	*Compliance service*	*Retrieves the rooms whose assessments were not satisfied, and their unsatisfied policy assessments (Use Case 2).*
3	*—*	*Reasoner*	*Finds the HVAC systems that serve those rooms.*
4	*—*	*Reasoner*	*Finds the components of those systems, including nested sub-components, that act on the failing measurement or on a measurement it was derived from.*
5	*—*	*Compliance service*	*Attaches to each candidate component the policy and the measurement that point to it. No cause is confirmed.*
6	*—*	*User interface*	*Returns the affected HVAC systems and the candidate components with their evidence.*
7	*System administrator*	*—*	*Inspects the candidate components first instead of the whole system.*

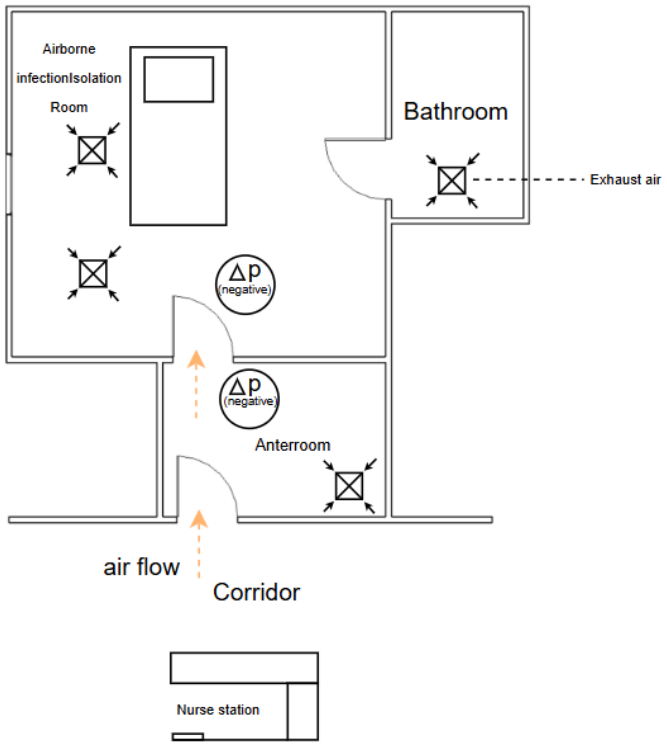
V. Alternate Flow of Events

Out-of-band condition: the evidence required for a determination is missing. This is the most common abnormal case in practice, and the behaviour is deliberately conservative - absence of evidence is never treated as evidence of compliance.

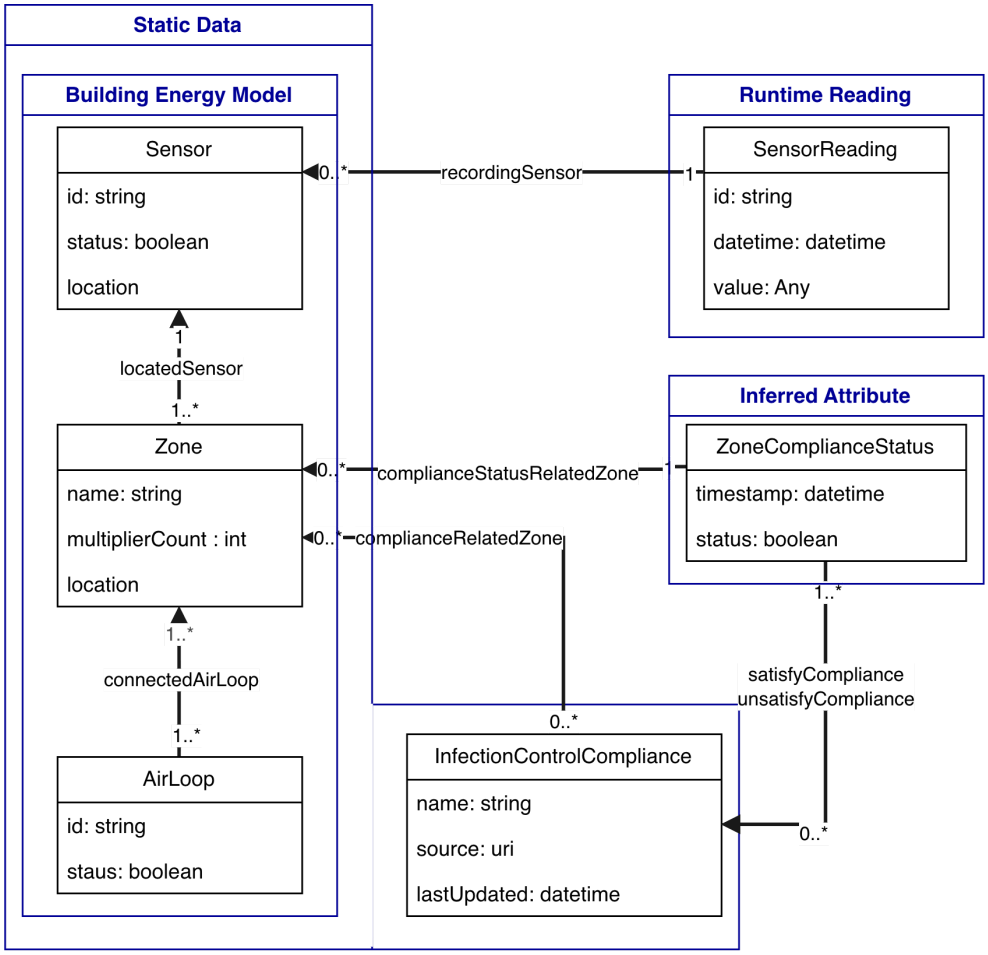
Alternate Flow of Events			
Step	***Actor (Person)***	***Actor (System)***	***Description***
4a	*—*	*Observation store*	*Cannot log a measurement for a policy assessment when asked (no sensor connected, or the sensor does not respond).*
4b	*—*	*Compliance service*	*Records the policy assessment as unknown. The room is never marked satisfied on the basis of an absent measurement.*
4c	*—*	*Compliance service*	*Records the reason: no sensor connected to the room, sensor not responding, or room covered by no framework.*
4d	*—*	*User interface*	*Surfaces unknown rooms separately and prominently rather than merging them with compliant ones.*
4e	*Healthcare administrator or nurse*	*—*	*Treats the room as not verified and asks facilities to check the sensor.*

VI. Use Case and Activity Diagram(s)

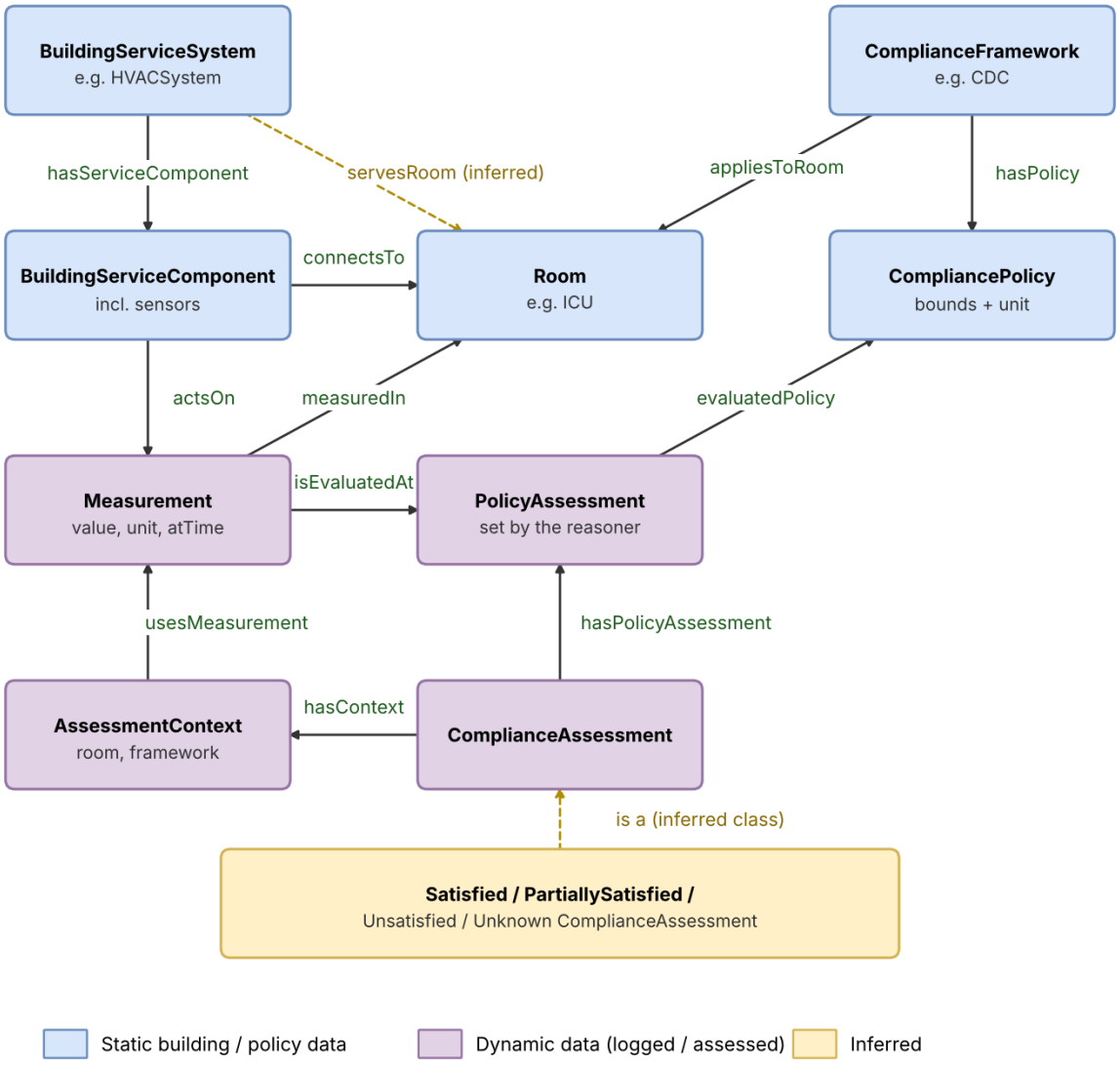




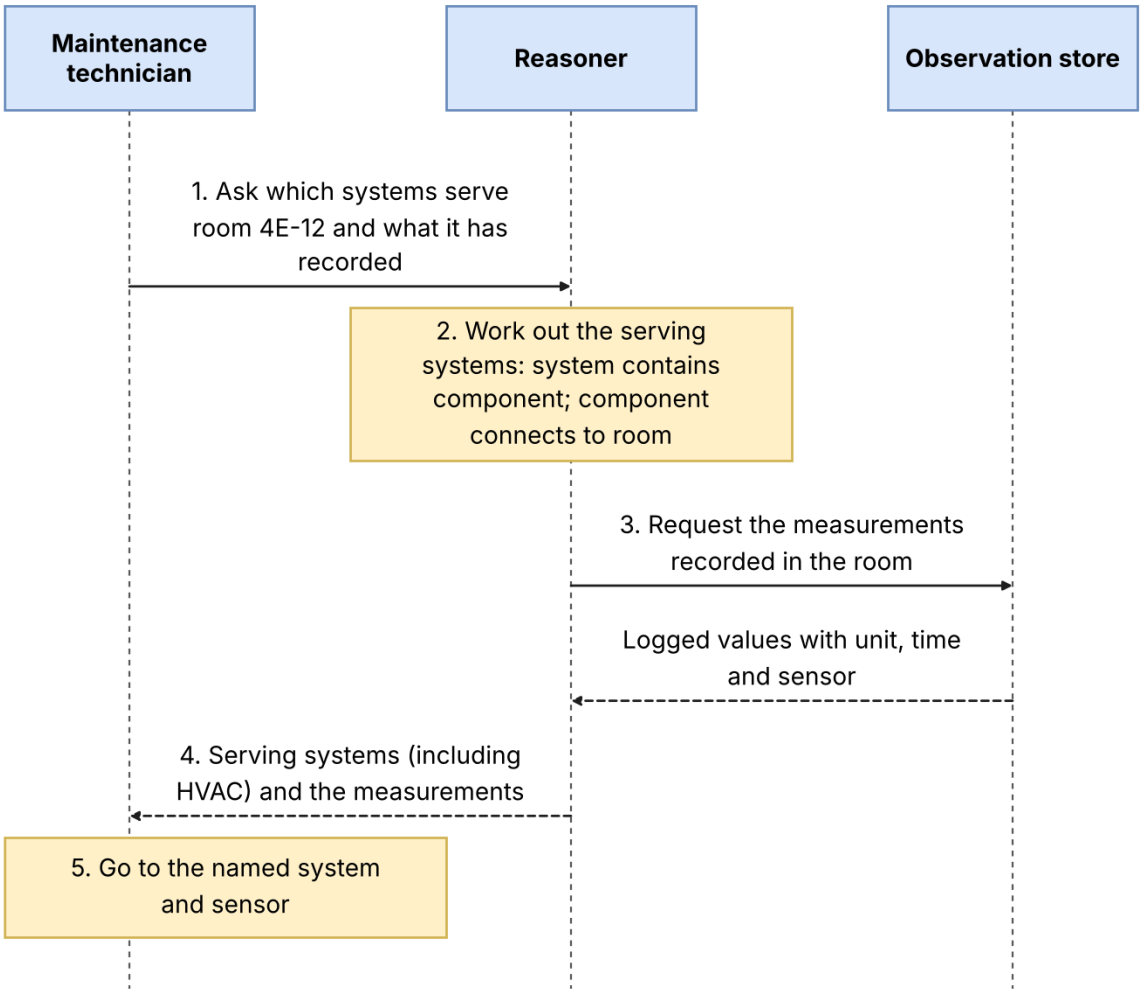
An instance of a hospital scenario



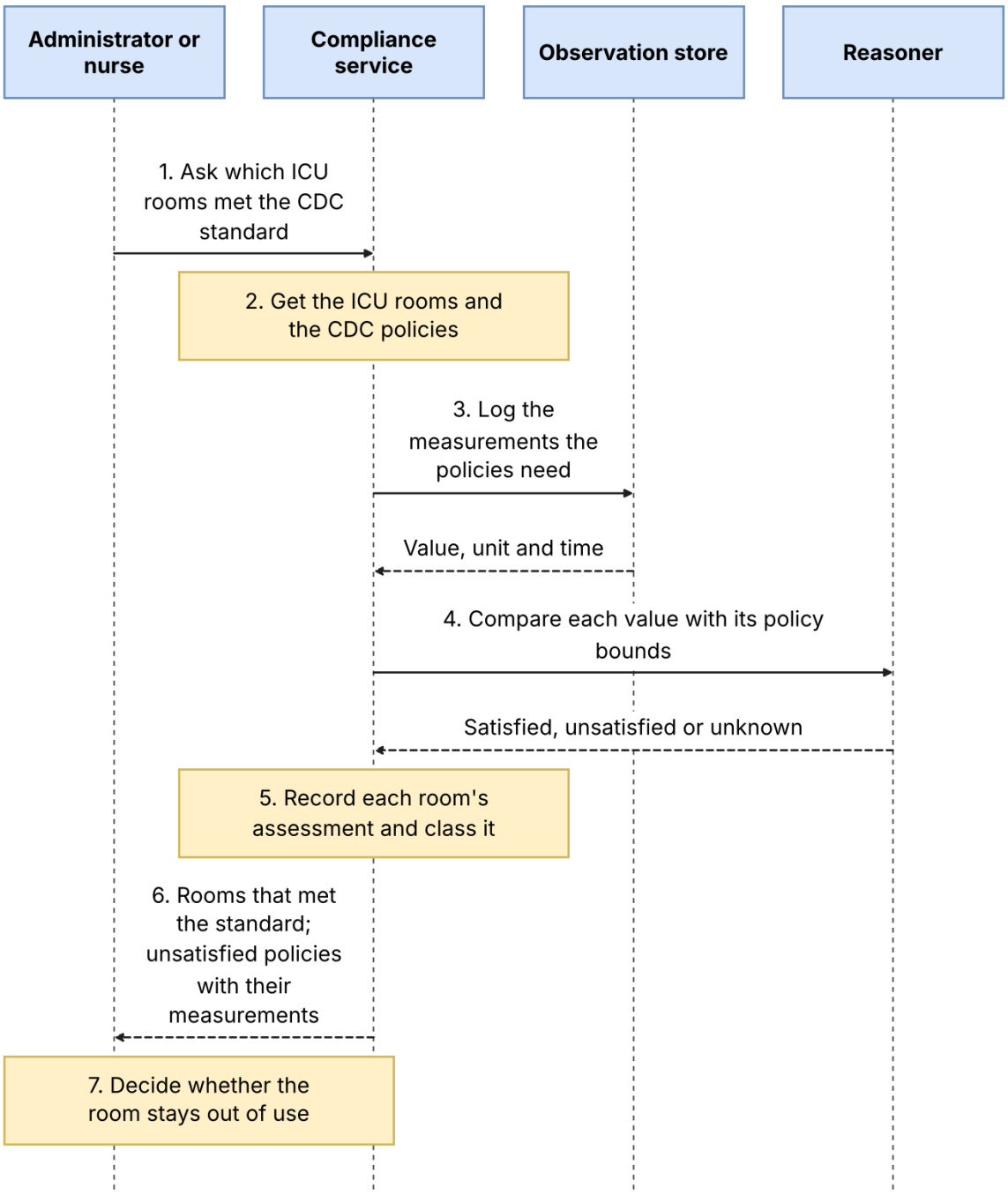
High-level architectural diagram



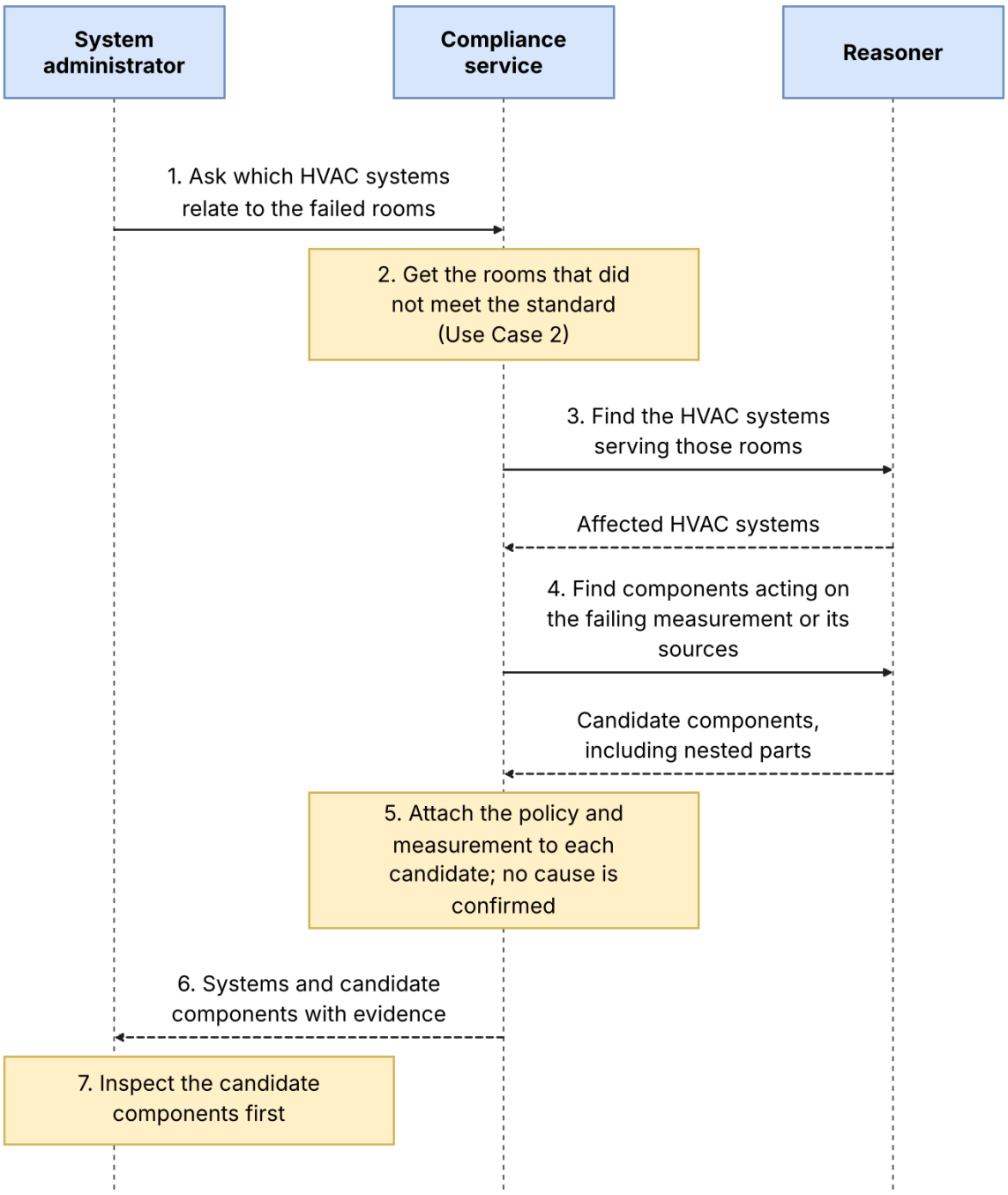
Simplified conceptual model of the system: building, policy, measurement and assessment data. Blue is static data, purple is dynamic data and yellow is inferred.



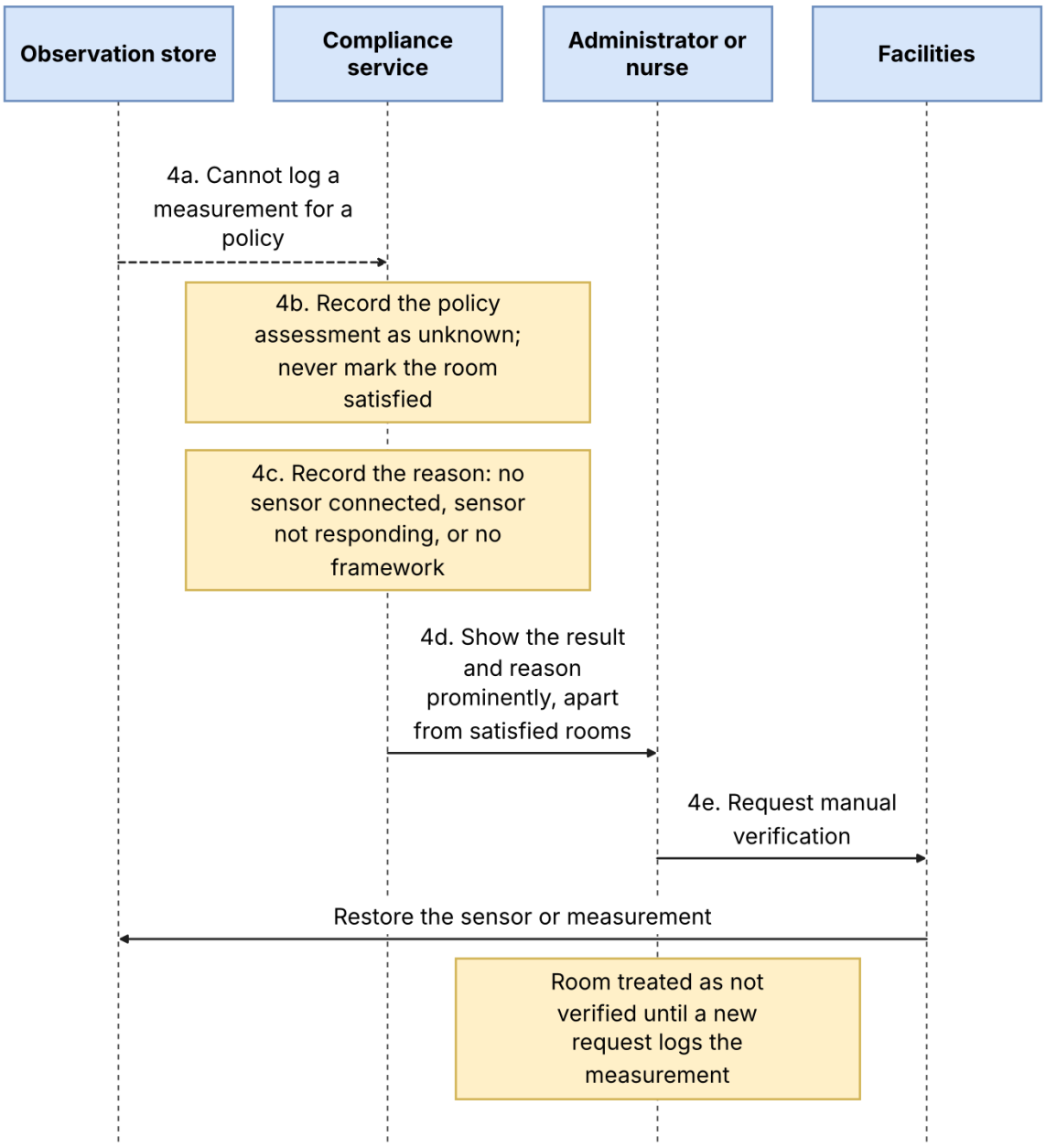
Sequence diagram for the Use Case 1 normal flow



Sequence diagram for the Use Case 2 normal flow



Sequence diagram for the Use Case 3 normal flow



Sequence diagram for the alternate flow

VII. Competency Questions

Focus of this revision: *Questions are grouped by use case: two per use case, six in total. The three use cases in focus are:*
★ ***Use Case 1:*** *CQ 1.2 - Which sensor readings have been recorded in room 4E-12, and which sensors produced them?*
★ ***Use Case 2:*** *CQ 2.2 - For room ICU-01, which compliance policies were not satisfied in that test, and which measurements show it?*
★ ***Use Case 3:*** *CQ 3.2 - Which components of HVAC-3 may be causing ICU-01 to fall short of the minimum outdoor air changes per hour policy?*
Each use case exercises a different kind of reasoning, which is why the set is not interchangeable. Use Case 1 works out relations that are never written down. Use Case 2 turns the comparison of measurements with

policy bounds into a compliance result for each room. Use Case 3 uses those results to trace failures back to the HVAC systems and components involved. Each question below gives the question, the example answer (only what a query returns), the ontology terms tested, how the ontology produces the answer, the reasoning in plain English, and the query in plain English. Example values are illustrative.
Reading the tags: *Ontology terms are tagged by the layer of the conceptual model they belong to. Shaded boxes are classes and terms with an arrow are relations or attributes. Blue is static building and policy data (`BuildingServiceSystem`, `→connectsTo`), green is a static relation that joins parts of the model (`→hasServiceComponent`), purple is dynamic data created by measuring and assessing (`SensorMeasurement`, `→hasContext`), and yellow is inferred, worked out from the other data and not stored (`SatisfiedComplianceAssessment`, `→servesRoom`). Instance names such as HVAC-3 and 4E-12 are plain text. An example answer contains only the answer, with the class of each answered item tagged; relations appear only in the explanations, where the answer is explained.*
Logging of measurements: *Sensor measurements are logged at request. A measurement is taken and recorded when a request asks for one; sensors are not polled on a schedule. Every answer below is therefore about the measurements logged so far, and when a needed measurement has not been logged the answer says unknown.*

Use Case 1 - Locating systems and sensor readings for a room

CQ 1.1 - Which building service systems serve room 4E-12?
Example answer: *Room 4E-12 `Room` is served by one building service system, the HVAC system HVAC-3 `HVACSystem`.*
Ontology terms tested: *Classes: `BuildingServiceSystem` `HVACSystem` `BuildingServiceComponent` `Room`. Relations: `→hasServiceComponent` `→connectsTo`, and `→servesRoom`, which is inferred.*
How the ontology produces this answer:

- *HVAC-3 `HVACSystem` contains `→hasServiceComponent` four components `BuildingServiceComponent`: the air handling unit AHU-3, the supply duct SD-3, the variable air volume box VAV-412 and the diffuser DF-412.*
- *VAV-412 and DF-412 each connect to `→connectsTo` room 4E-12. AHU-3 and SD-3 connect to 4E-12 as well, together with other rooms of the zone.*
- *An HVAC system is a kind of building service system `BuildingServiceSystem`, so a question about building service systems includes it.*

Reasoning:
The relation `→servesRoom` is inferred by a property chain, which is an OWL axiom and needs no extra rule: if a system has a service component, and that component connects to a room, then the system serves that room. Applied here, HVAC-3 has the service component VAV-412 and VAV-412 connects to 4E-12, so HVAC-3 serves 4E-12. The same chain fires through DF-412, AHU-3 and SD-3 and gives the same conclusion. A separate axiom handles nesting: having a service component is a kind of having a part, and having a part is transitive, so a part nested inside a component, such as a damper inside AHU-3, is also a part of HVAC-3. Such a nested part is a part of the system but is not used by the chain to conclude that the system serves a room. Because the relation is inferred and not stored, the answer changes when a component is added or re-connected, with no edit to any query.
Query (in plain English):
Find every building service system that serves room 4E-12, and say what kind of system it is (for example HVAC).

CQ 1.2 - Which sensor readings have been recorded in room 4E-12, and which sensors produced them?
Example answer: *Two measurements `SensorMeasurement` are recorded in room 4E-12 `Room`: 12.4 m³/h at 14:20 and 12.1 m³/h at 14:10, both produced by the air flow rate sensor AF-412 `AirFlowRateSensor`. They are the only readings logged so far.*
Ontology terms tested: *Classes: `SensorMeasurement` `AirFlowRateSensor` `ScalarQuantityValue` `Room`. Relations: `→measures` `→measuredIn` `→hasQuantityValue` `→hasMeasurementUnit` `→atTime`, and `→recordedIn`, which is inferred.*
How the ontology produces this answer:

- *When a measurement is logged at request, it is linked to the sensor that produced it `→measures` and*

to the room it was measured in `→measuredIn`*. It carries a value* `→hasQuantityValue` *with a unit* `→hasMeasurementUnit`*, and a time* `→atTime`*.*
- *The ontology states that being measured in a room is a more specific way of being recorded in it:* `→measuredIn` *is a sub-property of* `→recordedIn`*.*

Reasoning:
The relation `→recordedIn` *is inferred by sub-property reasoning, not by a property chain or a rule. Because* `→measuredIn` *is declared a sub-property of* `→recordedIn`*, every statement that a measurement was measured in 4E-12 also gives that it was recorded in 4E-12. The question asks for the general relation, so it returns every measurement recorded in the room, however the link to the room was stated, together with its sensor. Only measurements that have been logged are returned. The gap between 14:10 and 14:20 means that nobody asked for a reading in between, not that the sensor failed.*
Query (in plain English):
Find every measurement recorded in room 4E-12 and give its value, unit, time and the sensor that produced it.

Use Case 2 - Verifying ventilation compliance of critical care rooms

CQ 2.1 - Which ICU rooms met the CDC ventilation standard in the 14:30 test on 9 October 2026?
Example answer: *ICU-02 and ICU-03* `ICU` *met the CDC ventilation standard in the 14:30 test on 9 October 2026. ICU-01 did not.*
Ontology terms tested: *Classes:* `ICU` `CDC_compliance_framework` `ComplianceAssessment` `SatisfiedComplianceAssessment` `AssessmentContext` `PolicyAssessment` `SatisfiedPolicyAssessment` `Measurement`*. Relations:* `→appliesToRoom` `→hasPolicy` `→hasContext` `→forRoom` `→usesFramework` `→usesMeasurement` `→hasPolicyAssessment` `→evaluatedPolicy` `→isEvaluatedAt` `→atTime`*, and the membership in* `SatisfiedComplianceAssessment`*, which is inferred.*
How the ontology produces this answer:
- *ICU is a kind of room. The CDC framework applies to* `→appliesToRoom` *the three ICU rooms and has* `→hasPolicy` *six compliance policies.*
- *Each test of a room is recorded as a compliance assessment* `ComplianceAssessment`*. Its context* `→hasContext` `AssessmentContext` *names the room* `→forRoom`*, the framework* `→usesFramework` *and the measurements used* `→usesMeasurement`*, which were taken at 14:30* `→atTime`*.*
- *The assessment has one policy assessment* `→hasPolicyAssessment` *for each policy tested. Each names its policy* `→evaluatedPolicy` *and the measurement that was compared with it* `→isEvaluatedAt`*, and carries a state of satisfied, unsatisfied or unknown, recorded by the symbolic reasoner.*

Reasoning:
The answer comes from two separate steps. Step 1 is done by the symbolic compliance reasoner. It is not an OWL axiom and not a SWRL rule, because OWL class definitions cannot compare the numeric values of two different individuals. For each policy assessment the reasoner converts the measured value to the unit of the policy bound, then compares them: a policy with a lower bound is satisfied when the value is at least the bound, a policy with an upper bound when the value is at most the bound, and a range-bounded policy when both hold. It records the policy assessment as satisfied or unsatisfied, or as unknown when no measurement was logged or the units cannot be matched. Step 2 is inferred by the OWL reasoner from the ontology's definition: a compliance assessment is a `SatisfiedComplianceAssessment` *when it has at least one policy assessment and all of its policy assessments are satisfied. Applied here, every policy assessment of ICU-02 and ICU-03 was recorded as satisfied, so their assessments are classed as satisfied. ICU-01 has a policy assessment that was recorded as unsatisfied, so its assessment is not classed as satisfied (see CQ 2.2).*
Query (in plain English):
Find the ICU rooms whose compliance assessment from the 14:30 test on 9 October 2026 is classed as satisfied.

CQ 2.2 - For room ICU-01, which compliance policies were not satisfied in that test, and which measurements show it?
Example answer: *ICU-01* `ICU` *did not satisfy the minimum outdoor air changes per hour policy*

MinimumAirChangesOfOutdoorAirPerHour. *The policy requires at least 2 air changes per hour, and the outdoor air change rate* OutdoorAirChangePerHour *measured at 14:25 was 1.4. The other policies tested were satisfied.*

Ontology terms tested: *Classes:* PartiallySatisfiedComplianceAssessment UnsatisfiedPolicyAssessment MinimumAirChangesOfOutdoorAirPerHour OutdoorAirChangePerHour. *Relations:* →hasPolicyAssessment →evaluatedPolicy →hasLowerBound →hasUpperBound →evaluatesOn →isEvaluatedAt →atTime*, and the membership in* PartiallySatisfiedComplianceAssessment*, which is inferred.*

How the ontology produces this answer:

- *The assessment of ICU-01 has policy assessments* →hasPolicyAssessment*; the one for the minimum outdoor air changes policy was recorded as unsatisfied by the symbolic reasoner (CQ 2.1).*
- *That policy assessment names the policy it checked* →evaluatedPolicy*. The policy has a lower bound* →hasLowerBound *of 2 air changes per hour and states the kind of measurement it applies to* →evaluatesOn*.*
- *The measurement that was compared with it* →isEvaluatedAt *is the outdoor air change rate of 1.4, with its unit and its time* →atTime*.*

Reasoning:

One inference is made by the OWL reasoner. The assessment of ICU-01 has at least one satisfied and at least one unsatisfied policy assessment, so it is classed as a PartiallySatisfiedComplianceAssessment*, which tells the question that this room needs a closer look. The rest of the answer is stored, not inferred. The query follows the links from the assessment to its unsatisfied policy assessments, from each to its policy and the bound of that policy, and from each to the measurement compared with it. That measurement is the evidence. Policy assessments recorded as unknown are listed on their own, because no measurement was available for them and nothing can be said about the policy. The query looks at the policy assessments directly and does not rely on the class of the assessment, because an assessment that mixes unsatisfied and unknown policy assessments belongs to none of the four classes (an open point in Section II).*

Query (in plain English):

For room ICU-01 in the 14:30 test on 9 October 2026, find each policy assessment recorded as unsatisfied. For each, give the policy and its bound, and the measurement it was compared with, with value, unit and time. List the policy assessments recorded as unknown separately.

Use Case 3 - Tracing compliance failures to HVAC systems and components

CQ 3.1 - Which HVAC systems serve the rooms that did not meet the standard in the 14:30 test on 9 October 2026?

Example answer: *HVAC-3* HVACSystem *serves ICU-01* ICU*, which did not meet the standard. HVAC-4 serves only rooms that met the standard and is not returned.*

Ontology terms tested: *Classes:* ICU PartiallySatisfiedComplianceAssessment UnsatisfiedComplianceAssessment HVACSystem*. Relations:* →hasContext →forRoom →hasServiceComponent →connectsTo*, and* →servesRoom*, which is inferred.*

How the ontology produces this answer:

- *The rooms that did not meet the standard are the rooms named* →forRoom *in the context* →hasContext *of assessments that are classed as partially satisfied or unsatisfied (CQ 2.1 and CQ 2.2). Here that is ICU-01.*
- *The systems that serve those rooms come from the same inference as CQ 1.1: HVAC-3 has the service component VAV-412* →hasServiceComponent*, and VAV-412 connects to ICU-01* →connectsTo*.*
- *The result only shows that HVAC-3 is related to a failed room. A failed room does not by itself establish that the HVAC system is faulty.*

Reasoning:

The answer combines two inferences and adds no new axiom or rule. The first is from Use Case 2: the OWL reasoner classes each assessment as satisfied, partially satisfied, unsatisfied or unknown from its policy assessments, and the question keeps the rooms of the partially satisfied and unsatisfied ones. The second is from Use Case 1: the property chain from service component to connected room gives the →servesRoom *relation, which is followed from each failed room back to the systems. A room whose*

assessment is unknown, or mixes unknown with other states, is not returned (an open point in Section II).
Query (in plain English):
Find every HVAC system that serves at least one room whose compliance assessment from the 14:30 test on 9 October 2026 is classed as partially satisfied or unsatisfied, and list those rooms.

CQ 3.2 - Which components of HVAC-3 may be causing ICU-01 to fall short of the minimum outdoor air changes per hour policy?
Example answer: *Four components* `BuildingServiceComponent` *of HVAC-3* `HVACSystem` *are candidates: the variable air volume box VAV-412* `VariableAirVolumeBox`*, the diffuser DF-412* `Diffuser`*, the supply duct SD-3* `SupplyDuctwork` *and the filter FL-3* `Filter`*. Components of HVAC-3 that act only on measurements of policies that were satisfied are not returned.*
Ontology terms tested: *Classes:* `HVACSystem` `AirHandlingUnit` `VariableAirVolumeBox` `Diffuser` `SupplyDuctwork` `Filter` `OutdoorAirChangePerHour` `OutdoorAirFlowRateToRoom` `SupplyAirFlowRate`*. Relations:* `→evaluatesOn` `→wasDerivedFrom` `→actsOn` `→hasServiceComponent` `→isPartOf+`*, and the transitive reading of* `→isPartOf+`*, which is inferred.*
How the ontology produces this answer:

- *The policy that was not satisfied is evaluated on* `→evaluatesOn` *the outdoor air change rate. Its measurement, 1.4, is the failing measurement (CQ 2.2).*
- *That measurement was derived* `→wasDerivedFrom` *from the outdoor air flow rate to the room, which was derived from the supply air flow rate together with the outdoor air fraction and the room volume.*
- *Components are linked to the measurements they act on* `→actsOn`*. VAV-412, DF-412, SD-3 and FL-3 each act on the supply air flow rate.*
- *FL-3 is part of* `→isPartOf+` *the air handling unit AHU-3, and AHU-3 is a service component of HVAC-3* `→hasServiceComponent`*.*

Reasoning:
Three steps. First, the failing measurement is traced back to its sources. The relation `→wasDerivedFrom` *is not declared transitive in the ontology, so this is not an inference: the query follows it step by step, from the outdoor air change rate to the outdoor air flow rate to the room, and from there to the supply air flow rate, the outdoor air fraction and the room volume. Second, the components that act on any measurement in that trace are collected. Third, the OWL reasoner supplies the nesting:* `→isPartOf+` *is transitive and having a service component is a kind of having a part, so FL-3, which is part of AHU-3, is a part of HVAC-3. Only components of the systems found in CQ 3.1 are kept. Each is returned with the policy and the measurement that point to it, as a candidate and not as a confirmed cause.*
Query (in plain English):
For HVAC-3, find its components, including nested parts, that act on the outdoor air change rate that failed the policy or on any measurement it was derived from. Give each component with the policy and the measurement that point to it.

Coverage of the ontology: *The six questions use the room, service, measurement and compliance parts of the ontology. They do not use* `Building`*,* `BuildingSystem` *and the relations* `→hasBuildingSystem`*,* `→hasServiceSystem` *and* `→hasRoom`*, and they show only one of the six CDC policies in detail. Questions about a whole building, or about policies other than air changes, would cover them in the same pattern.*

Why semantics provides an advantage here?
The obvious alternative is a spreadsheet of rooms with their measured values and policy bounds, refreshed from the building management system. That approach can answer CQ 1.2, CQ 2.1 and CQ 2.2 for as long as the spreadsheet is maintained, and it cannot answer CQ 1.1, CQ 3.1 or CQ 3.2 at all, because the service relationship it would need is not recorded anywhere - it is a consequence of ductwork. Three specific advantages follow from an ontological representation. First, policies belong to a compliance framework rather than being copied per room, so applying the framework to a room automatically changes what is checked. Second, dependency is derived from connectivity, so the model survives changes to the mechanical system. Third, provenance is representable: a compliance claim can carry the observation,

timestamp and requirement clause that justify it, which is precisely what a regulatory surveyor asks for. A spreadsheet asserts that a room is compliant; the knowledge graph can show why.
That third advantage carries the energy consequence. ASHRAE 170 permits reduced air change rates in unoccupied spaces only on condition that the required pressure relationship is maintained and the full rate restored on occupancy. A facility that can only assert compliance has no defensible basis for taking that reduction and so ventilates at full rate continuously. A facility that can demonstrate compliance continuously, with provenance, does. The representation proposed here is therefore not only a safety instrument but the precondition for the most cost-effective ventilation energy measure available to a hospital.

VIII. Resources

Knowledge Bases, Repositories, or other Data Sources

Data	Type	Characteristics	Description	Owner	Source	Access Policies & Usage
DOE / PNNL Prototype Building Models - Hospital	*Remote, building model*	*EnergyPlus IDF / epJSON. Complete AirLoopHVAC definitions, zone volumes, spaces typed by healthcare function.*	*Primary building model for the project. Supplies the spaces, the zone volumes needed to convert airflow to air changes per hour, and the AHU to terminal unit to zone topology that CQ 1.1 and CQ 3.1 depend on.*	*U.S. Department of Energy, Building Energy Codes Program ; models developed by PNNL*	*https://www.energycodes.gov/prototype-building-models*	*OPEN - REQUIRED. US Government work, in the public domain. Freely downloadable and redistributable; no licence fee, no registration, no confidentiality restriction. Verified from the Building Energy Codes Program distribution page.*
CASE Office live sensor feed (Disruptive Technologies)	*Remote API, live*		*Working live sensor deployment already integrated with the author's IFC-to-223P pipeline. Provides real measurement data with device registry and units for prototyping before hospital data is available. Readings are requested from it and logged when needed, not collected by polling.*		*Disruptive Technologies REST API (institutional deployment)*	*REQUIRED for prototyping. Available to the author through CASE.*

External Ontologies, Vocabularies, or other Model Services

Resource	Language	Description	Owner	Source	Describes/Uses	Access Policies & Usage
ASHRAE Standard 223P	*OWL / RDF (Turtle)*	*Semantic model of building systems: equipment, connection points, connections, media, and observable properties. Reviewed as a source of topology terms. The current ontology does not import it.*		*https://open223.info*	*Describes the DOE Prototype Hospital topology and the sensor observations bound to it.*	*REQUIRED - core reuse. Openly accessible via open223.info. No licence assertion is present in the distributed Turtle file; we record that explicitly rather than assume a licence,*

						and will seek confirmation from ASHRAE before redistributing any derived artefact.
Brick Schema	*OWL / RDF*	*Ontology of building points, equipment, and locations. Used for typing sensor points where 223P is coarser.*		*https://brickschema.org*	*Describes sensor points; the model form used by Mortar.*	Reuse. BSD 3-Clause, asserted as dcterms:license within the distributed ontology file and verified there.
QUDT	*OWL / RDF*	*Reviewed for units. The current ontology uses the OMG Commons Quantities and Units vocabulary instead.*		*https://qudt.org*	*Reference only; units are taken from OMG Commons.*	*CC BY 4.0 , reference only*
SAREF / SAREF4BLDG	*OWL / RDF*	*ETSI smart-appliance and building ontologies, imported by the ontology for building service systems, components and sensors. Also reused by the prior OE indoor-environment-manager project.*		*https://saref.etsi.org*	*Describes the building service systems, components and sensors of the building model.*	ETSI Software License, Imported - reuse.
PROV-O	*OWL / RDF*	*W3C provenance ontology. Supplies the time of a measurement and the links between a derived measurement and its sources, and between an assessment and the systems and components it is associated with.*	W3C	*https://www.w3.org/TR/prov-o/*	*Used by measurements (time, derived from) and by assessments.*	W3C copyright notice and document-use rules. Imported - reuse.
REC (Real Estate Core)	*OWL / RDF*	*Ontology of buildings and rooms. Supplies the room classes, including the room types such as ICU, and the part-of relations between buildings and rooms.*	RealEstateCore	*https://w3id.org/rec*	*Describes the rooms and room types of the building model.*	MIT licence. Imported - reuse.
OMG Commons Quantities and Units	*OWL / RDF*	*Quantities and units. Every measurement value and every policy bound carries a unit so that the comparison is dimensionally sound.*	Object Management Group	*https://www.omg.org/spec/Commons*	*Used by every measurement and every policy bound.*	Copyright © 2026 Object Management Group, Imported - reuse.

Other Resources, Service, or Triggers *(e.g., event notification services, application services, etc.)*

Resource	Type	Description	Owner	Source	Access Policies & Usage
Differential pressure sensors	*Sensor*	*Report room measurements such as the room-to-corridor pressure differential and airflow. Primary evidence for CQ 1.2, CQ 2.1 and CQ 2.2.*		*Hospital BMS via BACnet; CASE deployment via IoT API*	*Assumed present in isolation rooms; required by code.*
SHACL validation engine	*Application service*	*Closed world reasoning specifically to assess policy assessment scenarios where assessment context doesn't have any measurement associated with.*	Project-hosted	*pySHACL or equivalent*	*Open source.*
Building management system / BACnet and IoT sensor API	*Sensor and control network service*	*Supplies differential pressure and airflow measurements, which are requested and logged on demand and not polled. Availability: not obtainable for the class project, so observations are simulated over the DOE Prototype Hospital at fixed*	*Hospital facilities department*	*BACnet/IP network; vendor BMS front end*	*Institutional and confidential. Not used in the class project; no hospital*

		timestamps.			*operational data is obtained, stored or published.*
SPARQL endpoint	*Query service*	*Executes the competency question queries, including the query that uses the derived serves relation.*	*Project-hosted*	*Apache Jena Fuseki or equivalent, deployed locally*	*OPEN source software, locally hosted. No access restriction.*
OWL reasoner	*Application service*	*Works out the inferred relations (serves a room, recorded in a room) and classes each compliance assessment as satisfied, partially satisfied, unsatisfied or unknown from its policy assessments.*	*Project-hosted*	*HermiT, Pellet or equivalent*	*Open source software, locally hosted. No access restriction.*

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X. Notes

Planned extensions:

The scope above is deliberately narrow for a first implementation. Each module is characterised below by the engineering technique it requires, because none of them requires prior knowledge of buildings.

- *Rule encoding (OWL, subsumption, taxonomy design).*

- *Topology and traversal (RDF, SPARQL property paths, graph traversal; depends on rule encoding for space identity).*
- *Validation under uncertainty (SHACL, SHACL-AF, three-valued evaluation).*
- *Explanation and provenance*
- *Natural-language interface (LLM prompting, NL-to-SPARQL, retrieval over graphs, benchmark evaluation)*
- *Additional space types (taxonomy extension). Protective environment rooms, sterile processing, bronchoscopy - each adding requirement kinds without changing the architecture.*

Acronyms and abbreviations

ABox – assertional component of a knowledge base - the instance data
ACH – air changes per hour
AHU – air-handling unit
AII – airborne infection isolation
ASHE – American Society for Health Care Engineering
ASHRAE – American Society of Heating, Refrigerating and Air-Conditioning Engineers
BACnet – Building Automation and Control Network protocol
BMS – building management system
CDC – Centers for Disease Control and Prevention
CQ – competency question
DOE – United States Department of Energy
FGI – Facility Guidelines Institute
HEPA – high-efficiency particulate air (filter)
ICRA – infection control risk assessment
ICU – intensive care unit
IDF – Input Data File - the EnergyPlus model format
IFC – Industry Foundation Classes
MERV – minimum efficiency reporting value - the filter efficiency scale used by Standard 170
OWL – Web Ontology Language
PE – protective environment
PNNL – Pacific Northwest National Laboratory
PROV-O – the W3C Provenance Ontology
QUDT – Quantities, Units, Dimensions and Types
RDF – Resource Description Framework
SHACL – Shapes Constraint Language
SPARQL – SPARQL Protocol and RDF Query Language
TAB – testing, adjusting and balancing
TBox – terminological component of a knowledge base - the ontology itself
UVGI – ultraviolet germicidal irradiation
VAV – variable air volume